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Using Statistical and Implied Volatility in Trading

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Defining Volatility

The term "volatility" is used in contexts ranging from financial markets and chemistry to geopolitics. Most traders in financial markets understand that volatility refers to measurements of the magnitude of price movements – or whipsaws, depending on one's perspective. The dictionary definition of volatile as it applies to financial markets is "tending to fluctuate sharply…"[1] or "tending to rapid and extreme fluctuations."[2]

Often misunderstood is that these references to volatility and price action are non-directional. To say that a market is showing "increased volatility" is not to say that it is making any directional progress, but rather only that prices are showing greater range or distribution.

Confusion on this point is compounded by discussions of volatility indexes such as the VIX, which is derived from the options markets. A rising VIX is often the product of concern over bearish price action that makes options traders willing to pay up for hedges (and drives others who may not otherwise trade options to seek such hedges); hence the association of a rising volatility index with downward price movement. Yet volatility itself is not a statement of direction.

Measuring Volatility

This Analysis Concepts paper discusses two common approaches to measuring the volatility for a specific asset. One approach uses the asset's price movements, while the other derives a volatility value from the asset's options premiums.

Statistical Volatility

Statistical volatility is any measure of volatility derived from the prices and price movements of the asset itself. TradeStation supplies two indicators of this type. Of these two, the indicator (formula) of interest to us is *Volatility Std Dev*. This uses the changes in closing prices over a number of bars to derive a volatility figure – specifically, one annual standard deviation expressed as a percent of price.

This approach is preferred here for several reasons.

First, it is based on the net changes in closing prices and disregards other prices from the bars. This is consistent with a traditional perspective on trend following and position trading in which indicators and signals are based on closing prices. That is the perspective we are adopting for this discussion.

Second, this expression of volatility can be used to compare assets since it is based on the distribution of closing prices and is expressed as a percent of price. The other indicator (formula), *Volatility*, uses bar ranges and so returns values expressed in the same terms as the asset, such that a high-priced asset will almost always have a higher value than a low-priced asset.

Third, the indicator *Volatility Std Dev* is consistent with expressions of volatility used in the options markets.

Implied Volatility

The advent of the options market and the financial engineering it has engendered afford a completely different tack on volatility. Since an option's value largely depends on the asset's volatility, options traders implicitly judge that volatility in assessing options premiums. More specifically and importantly, they express their expectation of the future volatility of the asset. After all, their positions and the premiums they pay or receive will pay off if the asset shows expected price action. One application of an options pricing model is to juggle the inputs and solve for the volatility component of the premiums.

> "This volatility would be the one that is being implied by the marketplace itself. That is, one only need ask the question, 'What volatility would I need to plug into the [options pricing] model to arrive at the current price of the option?' The answer is called the implied volatility."[3]

Note: Statistical volatility, by any measure, is often referred to as historical volatility. McMillan and others use this nomenclature. However, any measure of volatility that is reconstructed from historical market data – asset prices or options prices – is by definition historical. Therefore, in this paper we use the classifications statistical and implied, and values of either data series may be current or historical.

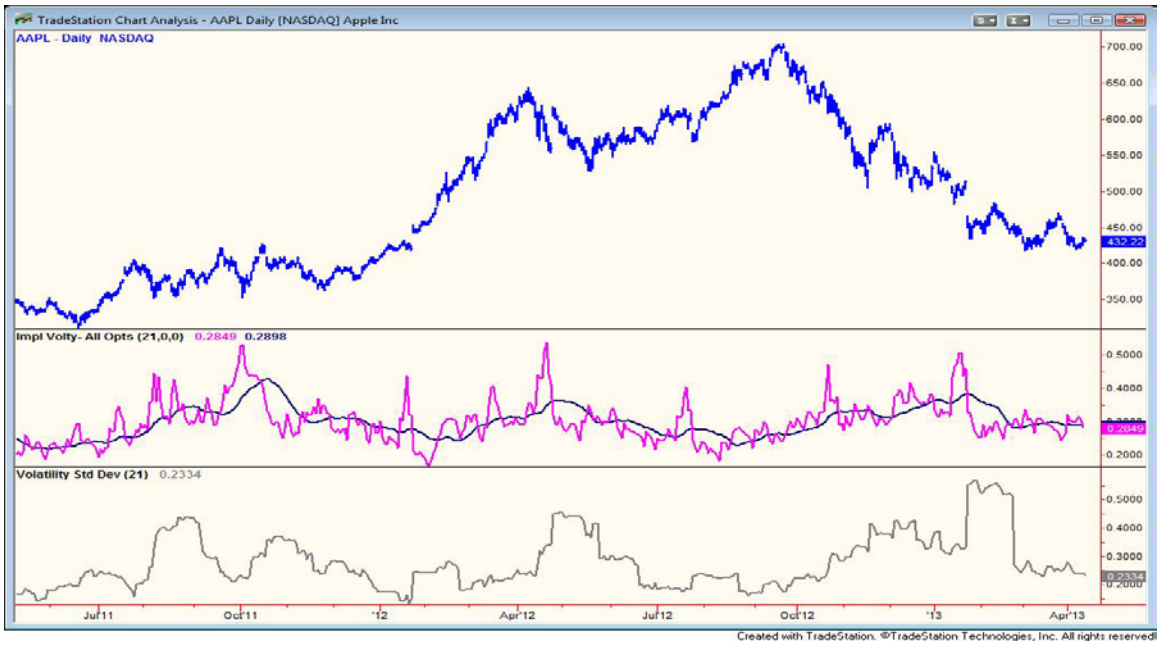
Options traders compare these two measures of volatility in search of options-trading opportunities afforded by seemingly mispriced options. Such mispricing may prompt them to modify a directional option position to exploit such mispricing, or may even give rise to market-neutral option positions designed to capitalize solely on such perceived mispricing.

The challenge undertaken here is not to assess methods of trading options using statistical and implied volatility comparisons. Instead, we examine the possibility of using these comparisons to profile the trending or non-trending nature of the asset. It is in the DNA of the options market to be a price discovery mechanism for volatility. How might this information be used in trading the asset?

Accessing and Charting Volatility Data

Figure 1 is a daily chart of Apple with two indicators, *Impl Volty- All Opts and Volatility Std Dev*. These two indicators are supplied by TradeStation.

Figure 1: Apple, daily



The magenta plot is the implied volatility calculated from Apple options. The black line is the 21-day simple moving average (SMA) of the implied volatility. That length is controlled by the input AvgLength. This input only affects the moving average of the implied volatility, not the implied volatility itself. (This input may be set to 1 to effectively hide the moving average.)

Beneath that, in gray, is the statistical volatility using the indicator *Volatility Std Dev*, calculated over the trailing 21 days. The input Length controls the number of days of data to use in the calculation.

Note: The implied volatility data field, and therefore the *Impl Volty - All Opts* indicator, may be applied only to daily and higher bar intervals; the *Volatility Std Dev* indicator (and underlying function) is designed to calculate properly on daily data only.

Comparing Statistical Volatility and Implied Volatility

This paper provides tools to examine the use of statistical volatility and implied volatility in identifying trending and non-trending periods. Do the leverage and pricing characteristics of options provide a filter for understanding expectations of price action? Other technical indicators must be used to identify either trend direction or range levels during non-trending periods.

The logic behind this notion is enumerated here.

1. Implied volatility is an expression of expectations. Therefore, when implied volatility is greater than statistical volatility, it may signal an expectation of upcoming price movement, and perhaps a move into a trending period.

2. Implied volatility, as shown in figure 1, is itself a volatile figure and so we smooth it using a simple moving average, also shown in figure 1. Twenty-one is used as the length of the average as a control; while other investigations should be made, this figure was chosen because it is the average number of business days in a month and between option expirations.

3. Under the theory that implied volatility has forward-looking characteristics, the moving average of the implied volatility is displaced forward. As a control, we chose 10 as the displacement length since it approximates half of 21. The comparison we will be making is between statistical volatility and the 21-day average of implied volatility of 10 days ago.

4. Statistical volatility is calculated using the same method as the indicator mentioned above and is shown in figure 1, *Volatility Std Dev*, with a length setting of 21, chosen as a control for the same reason as in point 2 above. Figure 2 has an indicator that simplifies and combines the statistical and implied volatility values of interest. The magenta plot is the 21-day average of implied volatility displaced forward 10 bars, per point 3 above. The gray line is the 21-day statistical volatility.

Figure 2 -- Apple, daily

5. The implementation of this is viewed as a filter for trend-following entries and may have less validity for exits. That is, while implied volatility may help identify upcoming trending price action, it often ebbs once the anticipated price action begins.

Figure 3 is the same price chart with an oscillator describing the relationship between implied and statistical volatility, per the points above. It is the difference between the two lines in the indicator in figure 2. In addition, a PaintBar study identifies those periods when the oscillator is above 0, or when implied volatility exceeds statistical volatility per our method above.

Figure 3 -- Apple, daily

Indicator -- TSL:IVol - SVol Oscillator

Plot	Description
IVOL-SVol	Plots the difference between the displaced average of implied volatility and statistical volatility, standard deviation.
0	Plots a horizontal reference line at 0

Input	Default	Description
IVLength	21	Length of simple average of implied volatility
IVOffset	10	Number of bars to offset simple average of implied volatility
SVLength	21	Number of bars to use in calculating statistical volatility, standard deviation
TrendingColor	Magenta	Plot color when implied volatility is greater than statistical volatility
NonTrendingColor	DarkBlue	Plot color when implied volatility is less than statistical volatility

PaintBar – TSL:IVol > SVol

Plot	Description
IVOL>SVol	Paints bars on which implied volatility is greater than statistical volatility

Input	Default	Description
IVLength	21	Length of simple average of implied volatility
IVOffset	10	Number of bars to offset simple average of implied volatility
SVLength	21	Number of bars to use in calculating statistical volatility, standard deviation

Sample Tests

A few simple steps are presented here to demonstrate the use of this data to profile a market as trending or non-trending and thereby filter trades. As noted above, the volatility calculations are non-directional so we must employ other indicators to use for trend signals to test with and without the volatility-based filter (implied volatility greater than statistical volatility). We created a test strategy employing two simple moving averages (SMA).

Strategy - TSL:IVol – SVol Trend Filter

Input	Default	Description
UseVolFilter	TRUE	Switch to turn on/off use of volatility filter
FastMALength	20	Length of fast (short) moving average
SlowMALength	50	Length of slow (long) moving average
IVLength	21	Length of simple average of implied volatility
IVOffset	10	Number of bars to offset simple average of implied volatility
SVLength	21	Number of bars to use in calculating statistical volatility, standard deviation

Signals	Description
Long Entry VF LE	Close greater than fast SMA and fast SMA greater than slow SMA, with volatility filter
Long Entry noVF LE	Close greater than fast SMA and fast SMA greater than slow SMA, without volatility filter
Short Entry VF SE	Close less than fast SMA and fast SMA less than slow SMA, with volatility filter
Short Entry noVF SE	Close less than fast SMA and fast SMA less than slow SMA, without volatility filter
Long Exit LX	Close less than slow SMA
Short Exit SX	Close greater than slow SMA

Note that the entry signals with and without the volatility filter are mutually exclusive. The use of the filter is governed by the input UseVolFilter.

Table 1 contains a composite summary of back-test results for this strategy applied to Apple, Inc. (AAPL). No commissions or slippage were taken out, and no optimization was done.

TradeStation Performance Summary		
Apple, Inc. Daily Bars - 5 years ending 3/28/13		
	Without Volatility Filter	With Volatility Filter
	All Trades	All Trades
Total Net Profit	$38,263.00	$45,049.00
Gross Profit	$64,708.00	$63,727.00
Gross Loss	($26,445.00)	($18,678.00)
Profit Factor	2.45	3.41
Total Number of Trades	34	30
Percent Profitable	23.53%	26.67%
Winning Trades	8	8
Losing Trades	26	22
Even Trades	0	0
Percent of Time in the Market	77.85%	73.63%

Simulated past performance does not guarantee future success.

The fields displayed were chosen to highlight the filtering possibilities of this volatility comparison, and to highlight how a filter should be examined. The overall effect on a controlled experiment is more important than an absolute result.

Profitability results, including Profit Factor – these results are improved by the use of the volatility filter.

Total Number of Trades and Losing Trades – the total number of trades is reduced, as would be expected with a signal filter. Note that all the trades that were filtered out were losing trades (22 vs. 26).

Percent of Time in the Market – this shows a small improvement.

Table 2 contains a composite summary of back-test results for this strategy on SPDR Gold Trust (GLD). No commissions or slippage were taken out, and no optimization was done.

TradeStation Performance Summary		
SPDR Gold Trust Daily Bars - 5 years ending 3/28/13		
	Without Volatility Filter	With Volatility Filter
	All Trades	All Trades
Total Net Profit	($3,210.00)	($1,038.00)
Gross Profit	$8,597.00	$8,944.00
Gross Loss	($11,807.00)	($9,982.00)
Profit Factor	0.73	0.9
Total Number of Trades	38	35
Percent Profitable	23.68%	25.71%
Winning Trades	9	9
Losing Trades	29	26
Even Trades	0	0
Percent of Time in the Market	77.91%	75.23%

Simulated past performance does not guarantee future success.

In this case, the strategy was not profitable either with or without the filter. Despite that, the comparison of significant fields highlights beneficial effects of the volatility filter.

Profitability results, including Profit Factor – these results are improved by the use of the volatility filter.

Total Number of Trades and Losing Trades – the same phenomenon is seen here as with Apple. The total number of trades is reduced, and all the trades that were filtered out were losing trades (26 vs. 29).

Percent of Time in the Market – this also shows a small improvement in this test.

Conclusion

Price discovery is a significant benefit of transparency in a market. The options market provides us with a price-discovery mechanism for volatility. This data and these calculations are available to traders of both options and their underlying assets.

This paper described the basics of implied volatility derived from the options for an asset and statistical volatility derived from price changes of that asset. An indicator and a PaintBar study to track the relationship are provided, along with a strategy to test the proposition that comparing statistical and implied volatility may be helpful in identifying trending and non-trending market periods.

Works Cited

1. *Webster's Encyclopedic Unabridged Dictionary* (Random House, 2001).
2. *Dictionary of Finance and Investment Terms* (Barron's, 1998).
3. McMillan, Lawrence. *Options as a Strategic Investment, 3rd Ed.* (New York: New York Institute of Finance, 1993), 728.

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